

As a **human computer interaction (HCI)** and **social computing** researcher, my work focuses on *understanding and reshaping how disruptive technologies (e.g. AI-mediated labor markets) interact with and affect the present and futures of workers*. To promote more collective, safe and prosperous worker communities, I draw on quantitative empirical methods (e.g. statistical analysis, LLM-supported interventions), grounded qualitative approaches (e.g. multi-stakeholder co-design/interviews in solidarity with workers) and system development (e.g. interaction prototypes).

One type of worker I study is **platform-based gig workers**, whose labor and services are increasingly woven into the fabric of everyday life. For consumers, gigs offer convenient, on-demand help with everyday tasks (from Ubering, to caretaking, to creative and professional projects) by drawing on the wide-ranging expertise of a global workforce. For workers, the temporary and task-based nature of gigs represents *independence* from traditional jobs and organizations – affording them the perceived freedom to choose not only *when* and *where* to labor¹, but also *what* to work on. However, as platforms mediate between consumers and workers, their hegemonic data practices shift significant agency and autonomy away from both stakeholder groups [a]. To centralize power, platforms capitalize on the absence of legal protections for independent contractors to exploit gig workers physically, financially and psychologically [b]. With rising rates of worker turnover and displacement, how do we create transformative interventions that propel workforces toward more natural, collective and liberated future interactions? Overall, the primary themes of my research:

- ① employs large-scale computational methods to identify worker practices [1] and intentions [11]
- ② co-designs with multi-stakeholder communities to envision collective, sustainable futures [2, 5, 6]
- ③ develops computational [7] interventions to facilitate more joyful [10], pluralistic [3, 9] and natural user interactions.

At the intersection of the HCI, CSCW² and design communities, my thesis work applied human-centered techniques to understand *interactions between disruptive technologies (i.e. gig platforms) and stakeholders (i.e. workers, consumers, regulators)*, followed by the **design and development of alternative interventions** to probe and reshape oppressive technology-mediated relations. I contributed to 12 publications/workshop (7 as first-author, 2 as advising author) across a range of venues: ACM CHI, CSCW, UIST, DIS, CHIWORK and IEEE VL/HCC. Two of my first-authored publications received best paper and honorable mention recognitions at CHI and CHIWORK, and I acquired over \$224,000 in funding as a NSF GRFP recipient and a Block Center for Technology and Society grant awardee.

By investigating timely challenges through multi-stakeholder and multi-scalar approaches, I aim to continue *surfacing timeless, generalizable designs and interventions* that guide future AI-mediated workforces in navigating disruptive technological change. I am passionate about creatively applying my empirical and design expertise to promote futures of work that (1) evolve individual human capabilities (2) motivate collective and supportive social relations and (3) cultivate deliberative, trusting and pluralistic communities.

① Understanding Worker Practices & Rationales via Large-Scale Data Analysis [1, 11]

Gig platforms commonly leverage a consumer-first rhetoric to justify and avoid accountability for unfair labor practices. But despite highly competitive, isolating platform environments, workers continue to willingly engage – suggesting the presence of underexamined intrinsic motivations that drive them to develop strategies for handling unexpected challenges. In my first pair of contributions, I present large-scale analyses that uncover (1) *how* freelancers communicate on platforms to achieve productivity and success (Fig. 1) as well as (2) crowdworker rationales on *why* they perform platform-mediated gigs despite harsh working conditions.

¹ For example, Uber marketed work flexibility through the slogan: “Drive when you want. Earn what you need.”

² Computer-supported cooperative work

The widespread adoption of remote work fueled tremendous growth for how virtually deliverable services (e.g. crowdsourcing, online freelancing) operate. But remote digital work ultimately lacks resources of established

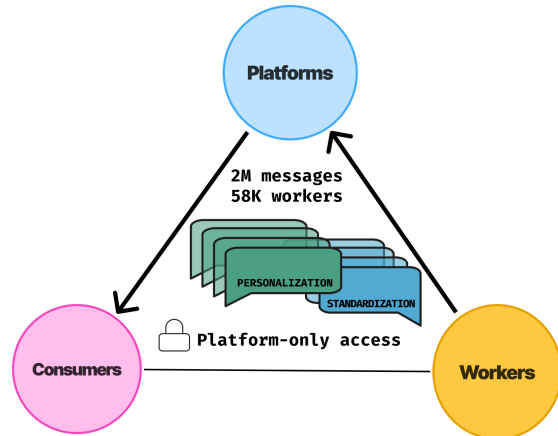


Fig 1. Platform-mediated Worker-Client Communication

workplaces and physically co-located peers, requiring workers to practice effective communication and time management. To **scalably understand individual worker strategies**, we engaged in multi-month negotiations with collaborators to access a large-scale dataset (> 2 million messages, >58k freelancers, >56k projects) from a leading online freelancing platform. Applying regression analyses on these platform-mediated freelancer-client communications, I found statistically significant correlations between (1) personalized message content and higher *job acquisition rates* (2) standardized messaging schedules and increased *job completion rates*, as well as (3) repeated uses of (standardized) templates across projects

that associated with greater bid volumes and *long term revenue* [1].

Understanding *how* workers achieve success on labor platforms reveals their strategies for reaching objectives. But *why* are workers motivated to continue engaging in online platform labor despite subpar conditions? Since many workers fear platform retaliation, we hypothesized that most workers' underlying beliefs about platforms remained undisclosed, under-examined, and likely misguided by platform-constructed illusions of flexibility. To explore latent worker *grievances* and *rationales*, we leveraged the persuasive power of language models to dissuade them from believing five common myths around gig work conditions. In our sample, 201 of 263 Prolific crowdworkers believed at least one myth. We collected these workers' rationales for a misguided belief and generated persuasive LLM responses to further probe their underlying incentives, revealing how **crowdworkers labor for platforms out of desires to preserve independence, autonomy and flexibility** [11]. Taken together, the two studies demonstrated feasibility of gathering actionable strategies from close-sourced platform data, as well as potential for provocative interventions to reveal hidden worker motivations.

② Co-designing Multi-Stakeholder Interventions for Platform Accountability [2, 5, 6, 7]

Mobilizing the data and efforts of online gig workforces revealed crucial strategies and rationales. But offline gig work requires support from higher-power stakeholders to enact infrastructural reforms that advance gig labor rights and protections.

To synergistically integrate priorities from multiple stakeholders, I led co-design workshops with policy experts, platform employees and workers to tackle five representative problems in gig work [2]. Our solution-oriented version of the speed-dating design method led stakeholders to generate and rank infrastructural interventions they are most prepared or motivated to implement (Fig. 2), which include visions of *stronger regulation*, *worker-owned cooperatives* and *platform-government co-regulation*. Stakeholders also envisioned more incremental advances such as platform-driven services/programs and collectivizing technologies (e.g. tools for enhancing scheduling, surfacing local resources and supporting financial well-being).

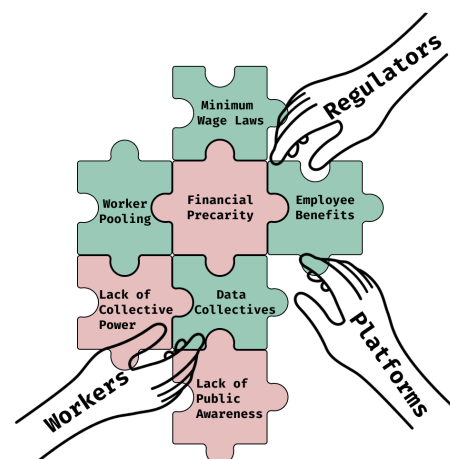


Fig. 2 Multi-Stakeholder Co-design of Solutions

Uncovering collective visions for infrastructural and regulatory change is a critical first step, but to achieve these in practice? In a CSCW '24 workshop, I led like-minded scholars to explore *worker data collectives* [6] for enacting counterdata production as well as advocacy and policy advances. But how to ensure that collected data actually support initiatives that matter? First I interviewed policy experts from varying levels of office to surface initiatives they prioritized to support with data-sharing (*i.e.* safety, fair pay) and system design requirements [5], which guided the architecture and interaction design for a prototype that I implemented with an interdisciplinary team. This culminated in [Gig2Gether](#) (Fig. 3), an open-sourced web app allowing workers to securely track and share gig work data across pink-, white- and blue-collar platforms, leverage visualization modules to view aggregated data and post qualitative stories about labor conditions. Together, these functions formed a complete data infrastructure for capturing and presenting scaled gig worker data to promote evidence-based advocacy and policymaking [6]. In a 7-day field evaluation with 16 participants, **Gig2Gether facilitated cross-platform mutual support** and financial reflection/planning, in addition to revealing latent worker desires for future use cases — including **collective algorithmic auditing, policy advocacy** as well as more complex interaction modes with fellow workers. For one Gig2Gether user, the tool represented “*a door between [gig workers] and politicians*”.

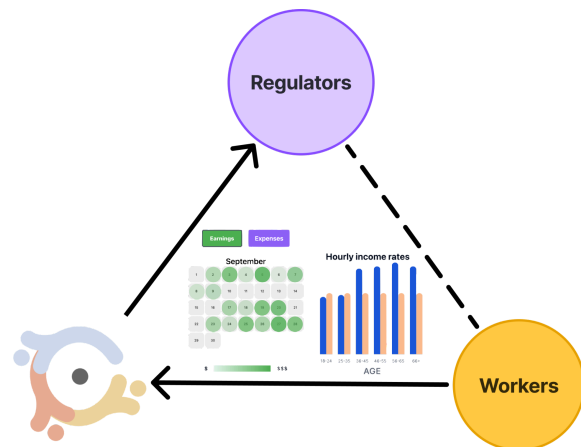


Fig 3. Gig2Gether Bridges Worker-Policymaker Data Gap

③ Building Social Solidarity & Pluralistic Interactions amidst Evolving Tech [3, 10]

Alongside technological interventions [2], regulators in our workshops also called for increased and individualized policy. In response, I synthesized legal and policy documents, labor regulations, exemplary state bills and HCI scholarship to envision advances. In an award-winning position paper [3], I advocated for targeted and pluralistic policies that consider diverse task domains and intersectional worker backgrounds.

While policymakers call for stronger labor regulations, platforms continuously rely on workers’ non-employee status to evade employer accountabilities – causing them to shoulder *vulnerabilities like immaterial labor and emotional burdens, alone* [c]. Meanwhile, HCI research has overlooked the role of consumers in worker advocacy – a gap that also harms workers when uninformed consumers hold unrealistic expectations of service quality. How to design socially approachable interventions to **motivate consumer-worker solidarity**?

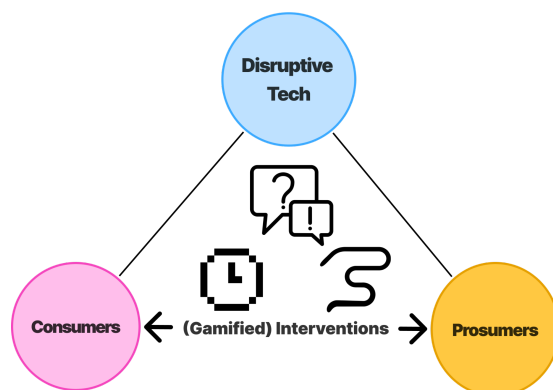


Fig 4. Promoting Consumer-driven Advocacy

To catalyze *worker-consumer co-liberation*, I introduced gamification as a novel alternative to data-sharing. For ridehail, I explored how in-ride games can spark joyful passenger engagement with latent driving conditions – *e.g.*, unseen labor, logistics and costs [10]. I led two students to develop six interactive games prototypes to probe stakeholder reactions and behaviors. Through workshops with 19 drivers and 15 riders, I uncovered desires for *natural and socially appropriate driver-passenger discussions of knowledge gaps* (*e.g.* long-term health impacts, rating-imposed service expectations) and design tradeoffs around managing content (*i.e.* lighter topics are less thought-provoking) and negotiations of consent – balancing passenger probing and driver sharing.

Future Research Aims

Gig platforms were disruptive technologies that transformed traditional relations between consumers and workers. I am eager to apply my expertise studying AI-mediated work with interdisciplinary, multi-stakeholder and scalable computational approaches to help emerging workforces *navigate transformative disruptions of rapidly-advancing technologies* as we approach more just, trusting and prosperous societies for all.

Designing Futures of Work that Capture Labor & Evolve *Individual Human Capabilities*. Outside labor relations, workers and consumers increasingly produce value and data for platforms as the creator economy grows. As AI permeates knowledge work domains, we must equip future workforces with tools that uplift, augment and transform their distinct human capabilities – e.g. intellect [d], creativity, empathy – when collaborating with AI. To support creative professionals, I advised an interdisciplinary (HCI+AI) team to design and evaluate POET, a tool that identifies and expands visual homogeneities of text-to-image models to *automatically generate more diverse outputs* [9]. This framework can extend to other modalities (e.g. music, video) and I am eager to explore alternative AI development that consider user agency in early-stage pipelines.

Besides creativity, emergent interactive and sensing modalities such as video and speech offer exciting pathways for capturing tacit knowledge, invisible labor and latent human expertise. Currently, the high effort required to document hidden costs and unpaid work prevents and demotivates workers (especially those performing more service-based, low-income and vulnerable forms of work) from actively engaging in peer data-sharing. Combining empirical insights from labor studies and social/organizational science with technical possibilities from data science and engineering (e.g., embedded design, data visualization, graphics), I aim to develop richer, more effortless collection and interaction mechanisms to (1) help workers gather and structure evidence to *visibilize and value labor*; and (2) *amplify human knowledge and capabilities* more broadly.

Reorienting *Social Relations* to Motivate Co-Liberating Collective Actions. Gamification [10] represented one technique to incentivize consumer engagement and solidarity with workers. Another way to promote worker-consumer unity is to bring to users' attention that *gig workers are also consumers* who benefit from platform resources – e.g. their client pools, matching algorithms. Notably, paying consumers hold significant collective power over platform decisions, and I will apply frameworks from organizational management and critical computing to analyze, consider and build mechanisms/relations to facilitate bottom-up accountability.

Beyond gig labor markets, union membership in general is declining (currently <10% of US adults [e]), while labor regulations and protections are falling behind rapidly-evolving technologies and worker needs. As growing segments of the workforce upskill and adapt to AI-assisted and AI-mediated workflows, I will investigate multi-stakeholder relations to explore novel interventions – leveraging creative and novel techniques such as behavioral economics, game or mechanism design – to motivate and foster solidarity among workers and their allies. More *scalably*, I will employ observational and quasi-experimental data analyses to gather and measure actionable insights that align more closely with worker (career) objectives.

Building Deliberative, Pluralistic and Trust-based *Communities*. Online communities of today (e.g. social media, open source development) suffer from misaligned incentives and disruption – e.g. inappropriate comments [4]. Sustainable communities depend on trust, which requires shared understandings of on-the-ground realities and negotiations to reconcile pluralistic values and objectives. PolicyCraft showed how human and AI-generated cases facilitated deliberation in education [8]. Many emerging AI capabilities carry potential to transform our understanding and trust for each other (e.g. persuasion [11]). For instance, agent-based simulations offer scalable opportunities to develop more natural, unremarkable [f] software systems that promote prosocial and deliberative communities and societies. I intend to start with familiar domains (e.g. gig work, art [12]) before exploring newer realms – e.g. music, game design [10] world-building.

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